CBSE Class 12 Chemistry — Half-Yearly Predicted Paper (Set 7 of 10)

50 Most-Probable PYQs & Board-Pattern Practice Questions

Section A: 1-Mark Questions (MCQs / Assertion-Reason)

Q1. Blood cells are isotonic with $0.9\%(w/v)$ sodium chloride solution. What happens if human blood cells are placed in a 0.4% sodium chloride solution? (a) They will shrink due to plasmolysis. (b) They will swell and may even burst due to endosmosis. (c) They will remain perfectly unchanged. (d) They will undergo rapid coagulation. [Solutions] [PYQ 2016] [Confidence: Very High]

Q2. For a chemical reaction, the plot of $\ln k$ versus $1/T$ (where k is the rate constant and T is absolute temperature) gives a straight line. The slope of this line is equal to: (a) $\frac{E_a}{R}$ (b) $\frac{-E_a}{R}$ (c) $\frac{E_a}{2.303R}$ (d) $\frac{-E_a}{2.303R}$ [Chemical Kinetics] [PYQ 2020] [Confidence: Very High]

Q3. During the electrolysis of an aqueous solution of $CuSO_4$ using inert platinum (Pt) electrodes, the product liberated at the anode is: (a) Copper metal (b) Hydrogen gas (c) Oxygen gas (d) Sulphur dioxide gas [Electrochemistry] [Practice] [Confidence: High]

Q4. Which of the following transition elements of the 3d series exhibits the lowest enthalpy of atomization? (a) Scandium (Sc) (b) Manganese (Mn) (c) Zinc (Zn) (d) Copper (Cu) [The d- and f-Block Elements] [PYQ 2019] [Confidence: Very High]

Q5. The correct IUPAC formula for the complex 'Potassium trioxalatochromate(III)' is: (a) $K_3[Cr(C_2O_4)_3]$ (b) $K[Cr(C_2O_4)_3]$ (c) $K_2[Cr(C_2O_4)_3]$ (d) $K_4[Cr(C_2O_4)_3]$ [Coordination Compounds] [PYQ 2015] [Confidence: Very High]

Q6. The reaction of ethyl bromide (CH_3CH_2Br) with potassium nitrite (KNO_2) predominantly yields: (a) Nitroethane (b) Ethyl nitrite (c) Ethyl cyanide (d) Ethanol [Haloalkanes and Haloarenes] [PYQ Theme] [Confidence: High]

Q7. When tertiary butyl alcohol, $(CH_3)_3C-OH$, is passed over heated copper at $573\ K$, the major organic product formed is: (a) Butan-2-one (b) Isobutylene (2-Methylpropene) (c) Isobutane (d) 2-Methylpropanal [Alcohols, Phenols and Ethers] [PYQ 2020] [Confidence: Very High]

Q8. The specific reagent mixture used in the Stephen reduction to convert a nitrile to an aldehyde is: (a) $SnCl_2 + HCl$ (b) $Zn(Hg) + \text{conc. } HCl$ (c) $LiAlH_4$ (d) $H_2/Pd - BaSO_4$ [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2015] [Confidence: Very High]

Q9. The molal depression constant (K_f) of a solvent depends strictly upon: (a) The nature of the solute dissolved in it (b) The nature of the solvent (c) The vapour pressure of the solution (d) The molality of the solution [Solutions] [PYQ 2021] [Confidence: Very High]

Q10. The highest known oxidation state exhibited by any transition element is +8. Which of the following metals exhibits this state? (a) Manganese (Mn) (b) Osmium (Os) (c) Chromium (Cr) (d) Iron (Fe) [The d- and f-Block Elements] [Practice] [Confidence: High]

Q11. For a reaction $A + B \rightarrow \text{Products}$, the rate doubles when the concentration of B is doubled. When the concentrations of both A and B are doubled, the rate increases by a factor of 8. The rate law for the reaction is: (a) $Rate = k[A][B]^2$ (b) $Rate = k[A]^2[B]^2$ (c) $Rate = k[A]^2[B]$ (d) $Rate = k[A][B]$ [Chemical Kinetics] [PYQ 2016] [Confidence: Very High]

Q12. The nucleophilic addition of Hydrogen Cyanide (HCN) to benzaldehyde yields: (a) Benzyl alcohol (b) Benzaldehyde cyanohydrin (c) Benzoic acid (d) Phenylacetonitrile [Aldehydes, Ketones and Carboxylic Acids]

[PYQ 2014] [Confidence: Very High]

Directions for Q13 to Q16: A statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q13. Assertion (A): Primary alkyl halides are generally more reactive than secondary and tertiary alkyl halides towards an S_N2 reaction. Reason (R): The S_N2 mechanism exclusively involves the inversion of configuration of the carbon atom. [Haloalkanes and Haloarenes] [Practice] [Confidence: High]

Q14. Assertion (A): The boiling point of $0.1\ M\ MgCl_2(aq)$ is noticeably higher than that of $0.1\ M\ NaCl(aq)$. Reason (R): Assuming complete dissociation, $MgCl_2$ furnishes more ions (3 moles of ions per mole) in solution than $NaCl$ (2 moles of ions per mole). [Solutions] [PYQ Theme] [Confidence: Very High]

Q15. Assertion (A): The complex ion $[Fe(H_2O)_6]^{3+}$ is an outer orbital complex. Reason (R): Water (H_2O) is a weak field ligand and cannot force the pairing of electrons in the 3d orbitals of Fe^{3+}. [Coordination Compounds] [PYQ 2016] [Confidence: Very High]

Q16. Assertion (A): p-Nitrophenol is a significantly stronger acid than o-nitrophenol. Reason (R): Intra-molecular hydrogen bonding in o-nitrophenol traps the hydrogen atom, making the loss of a proton slightly more difficult than in p-nitrophenol. [Alcohols, Phenols and Ethers] [PYQ 2020] [Confidence: Very High]

Section B: 2-Mark Questions (Short Answer)

Q17. Define the term *Reverse Osmosis*. Write one crucial practical application of reverse osmosis. [Solutions] [PYQ 2019] [Confidence: Very High]

Q18. A lead storage battery is a highly useful secondary cell. Write the overall balanced chemical reaction that occurs when this battery is being recharged by passing a current through it. [Electrochemistry] [PYQ 2020] [Confidence: High]

Q19. Define "Activation Energy" (E_a) and "Threshold Energy" in chemical kinetics. How are they mathematically related to each other? [Chemical Kinetics] [Practice] [Confidence: High]

Q20. Account for the following strict observation in d-block chemistry: The copper(I) ion (Cu^+) is not stable in an aqueous solution and readily undergoes disproportionation. [The d- and f-Block Elements] [PYQ 2023] [Confidence: Very High]

Q21. Predict the structures of the organic products A and B in the following sequence of reactions: $CH_3 - C \equiv CH \xrightarrow{H_2O,\ H_2SO_4,\ HgSO_4} A \xrightarrow{I_2,\ NaOH} B + CHI_3 \downarrow$ [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2016] [Confidence: Very High]

Section C: 3-Mark Questions (Short Answer)

Q22. Give distinct chemical reasons to account for the following observations: (i) The $C - Cl$ bond length in chlorobenzene is notably shorter than that in chloromethane (CH_3Cl). (ii) A true racemic mixture is optically inactive. (iii) Grignard reagents ($RMgX$) must be prepared strictly under anhydrous conditions. [Haloalkanes and Haloarenes] [PYQ 2017 / 2018] [Confidence: Very High]

Q23. Write the structures of the major organic product(s) formed in each of the following reactions: (i) Propene is treated with B_2H_6 followed by oxidation with H_2O_2/OH^-. (ii) Phenol is treated with concentrated HNO_3 in the presence of concentrated H_2SO_4. (iii) Anisole is reacted with CH_3Cl in the presence of anhydrous $AlCl_3$. [Alcohols, Phenols and Ethers] [PYQ 2019 / 2022] [Confidence: Very High]

Q24. A solution containing $1.9\ g$ of Magnesium Chloride ($MgCl_2$, Molar mass = $95\ g\ mol^{-1}$) dissolved in $50\ g$ of water freezes at $-1.95°C$. Determine the exact degree of dissociation (α) of $MgCl_2$ in this solution. (Given: K_f for water = $1.86\ K\ kg\ mol^{-1}$). [Solutions] [PYQ 2016] [Confidence: Very High]

Q25. The molar conductivity of a $0.025\ mol\ L^{-1}$ aqueous solution of methanoic acid ($HCOOH$) is $46.1\ S\ cm^2\ mol^{-1}$. Calculate its degree of dissociation (α) and its dissociation constant (K_a). (Given: $\lambda°(H^+) = 349.6\ S\ cm^2\ mol^{-1}$ and $\lambda°(HCOO^-) = 54.6\ S\ cm^2\ mol^{-1}$). [Electrochemistry] [PYQ 2015] [Confidence: Very High]

Q26. (i) Draw the two geometrical isomers of the coordination complex $[Co(en)_2Cl_2]^+$. (ii) Out of these two isomers, strictly identify which one is optically active and state the reason why. [Coordination Compounds] [PYQ 2016] [Confidence: Very High]

Q27. Give simple, observable chemical tests to distinctly differentiate between the following pairs of compounds: (i) Benzaldehyde and Propanal (ii) Phenol and Cyclohexanol (iii) Diethyl ether and Ethanol [Aldehydes, Ketones and Carboxylic Acids / Alcohols] [PYQ 2014 / Practice] [Confidence: High]

Q28. Complete and balance the following crucial ionic equations from d-block chemistry: (i) $Cr_2O_7^{2-} + Fe^{2+} + H^+ \rightarrow$ (ii) $MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow$ (iii) $MnO_4^- + H_2O + I^- \rightarrow$ [The d- and f-Block Elements] [PYQ 2018 / 2019] [Confidence: Very High]

Section D: 4-Mark Questions (Case-Study)

Q29. Case Study: Kinetics of Radioactive Decay and First-Order Reactions The rate of a first-order reaction depends strictly on the first power of the concentration of the reactant. Natural and artificial radioactive decay of unstable nuclei elegantly follow first-order kinetics. For instance, Carbon-14 dating, which is heavily used in archaeology to determine the age of ancient biological artifacts, relies entirely on the steady first-order decay rate of the $C-14$ isotope. The time taken for exactly half of the initial concentration of a reactant to decay is called its half-life ($t_{1/2}$), which for a first-order process is entirely independent of the starting amount. (a) State the unit of the specific rate constant (k) for a first-order reaction. (1 Mark) (b) Mathematically, what exact fraction of the original reactant will be left unreacted after 3 half-lives have passed? (1 Mark) (c) A specific first-order reaction is found to be exactly 20% complete in 10 minutes. Calculate the specific rate constant (k) of this reaction. (Given: $\log 10 = 1.000, \log 8 = 0.903$). (2 Marks) [Chemical Kinetics] [Practice based on PYQ Themes] [Confidence: Very High]

Q30. Case Study: Ethers and Williamson Synthesis Ethers are significantly less polar than alcohols and completely lack intermolecular hydrogen bonding, making them highly volatile. The Williamson synthesis is an excellent and versatile laboratory method for the preparation of both symmetrical and unsymmetrical ethers. It involves the S_N2 nucleophilic attack of a sodium alkoxide ion on a primary alkyl halide. However, because alkoxides are not only strong nucleophiles but also remarkably strong bases, using secondary or tertiary alkyl halides completely alters the reaction pathway, leading heavily to elimination products (alkenes) rather than substitution products. (a) Write the structures of the exact products formed when tert-butyl methyl ether, $(CH_3)_3C-O-CH_3$, reacts with exactly one equivalent of hot Hydrogen Iodide (HI). (1 Mark) (b) Why cannot bromobenzene be reliably used in Williamson's synthesis to prepare phenetole (ethyl phenyl ether)? (1 Mark) (c) Write the suitable exact reactants (the specific alkyl halide and the specific sodium alkoxide) needed to successfully prepare 2-methoxy-2-methylpropane. Explain briefly what would happen if their roles were reversed. (2 Marks) [Alcohols, Phenols and Ethers] [PYQ 2015 / 2020] [Confidence: High]

Section E: 5-Mark Questions (Long Answer)

Q31. (a) State Kohlrausch's law of independent migration of ions. Write the exact mathematical expression for the limiting molar conductivity (Λ_m°) of Aluminium sulphate, $Al_2(SO_4)_3$, using this law. (2 Marks) (b)

Calculate the e.m.f. of the following galvanic cell at $298\ K$: $Fe(s)|Fe^{2+}(0.001\ M)||H^+(1\ M)|H_2(g)(1\ \text{bar})|Pt(s)$ (Given: Standard electrode potential $E^\circ_{Fe^{2+}/Fe} = -0.44\ V$. Assume $\frac{2.303RT}{F} = 0.0591\ V$). (3 Marks) [Electrochemistry] [PYQ 2015 / 2019] [Confidence: Very High]

Q32. (a) Transition elements and their coordination compounds are widely used as catalysts in industrial processes. Give two strong chemical reasons to justify this capability. (2 Marks) (b) Using Valence Bond Theory (VBT), clearly explain the state of hybridization, the resulting geometry, and the magnetic behaviour of the complex ion $[NiCl_4]^{2-}$. (Given: Atomic number of Ni = 28). (3 Marks) [The d- and f-Block Elements / Coordination Compounds] [PYQ 2017 / 2014] [Confidence: Very High]

Q33. (a) An organic compound **A** (molecular formula $C_8H_{16}O_2$) was hydrolysed with dilute sulphuric acid to give a carboxylic acid **B** and a straight-chain alcohol **C**. Vigorous oxidation of the alcohol **C** with chromic acid produced the exact same carboxylic acid **B**. Identify the structures and IUPAC names of compounds **A**, **B**, and **C**, and write the balanced chemical equations for the hydrolysis and oxidation reactions involved. (3 Marks) (b) Describe the following important named organic reactions with a suitable balanced chemical equation for each: (i) Hell-Volhard-Zelinsky (HVZ) reaction (ii) Decarboxylation of a carboxylic acid (2 Marks) [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2016 / 2018] [Confidence: Very High]

Answer Key

Q1. (b) They will swell and may even burst due to endosmosis. (Water moves from the hypotonic 0.4% solution into the hypertonic blood cells). **Q2.** (b) $\frac{-E_a}{R}$. (From Arrhenius eq: $\ln k = \ln A - \frac{E_a}{RT}$). **Q3.** (c) Oxygen gas. (At the inert anode, water is oxidized in preference to sulphate ions: $2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$). **Q4.** (c) Zinc ($Zn$). (It has completely filled 3d orbitals ($3d^{10}$), meaning no unpaired d-electrons are available to form strong metallic bonds). **Q5.** (a) $K_3[Cr(C_2O_4)_3]$. (Oxalate is -2. Cr is +3. Total charge of complex sphere = $3 + 3(-2) = -3$. So 3 K^+ are needed). **Q6.** (b) Ethyl nitrite. (KNO_2 is an ionic compound and attacks through the oxygen atom, yielding an alkyl nitrite. $AgNO_2$ yields nitroalkane). **Q7.** (b) Isobutylene (2-Methylpropene). (Tertiary alcohols undergo dehydration instead of dehydrogenation/oxidation when passed over heated copper). **Q8.** (a) $SnCl_2 + HCl$. **Q9.** (b) The nature of the solvent. (It is a constant specific to the solvent, $K_f = \frac{R \cdot M_1 \cdot T_f^2}{1000 \cdot \Delta_{fus} H}$). **Q10.** (b) Osmium ($Os$). (Ruthenium also exhibits +8). **Q11.** (c) $Rate = k[A]^2[B]$. (Doubling B doubles rate $\implies$ 1st order in B. Doubling A & B increases rate $8\times$; since B accounts for $2\times$, A must account for $4\times$, so 2nd order in A). **Q12.** (b) Benzaldehyde cyanohydrin. (Nucleophilic addition of CN^- followed by protonation). **Q13.** (b) Both A and R are true but R is not the correct explanation of A. (Primary halides are more reactive because of less steric hindrance around the carbon atom, not just because inversion occurs). **Q14.** (a) Both A and R are true and R is the correct explanation of A. **Q15.** (a) Both A and R are true and R is the correct explanation of A. **Q16.** (a) Both A and R are true and R is the correct explanation of A.

Q17. If a pressure higher than the osmotic pressure is applied to the solution side, the flow of solvent is reversed, passing from the solution into the pure solvent. Application: Desalination of sea water to obtain fresh drinking water. **Q18.** Recharging reaction: $2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$. **Q19.** Activation Energy (E_a): The minimum extra amount of energy required by reacting molecules to form the activated complex. Threshold Energy: The minimum total kinetic energy the molecules must possess to react. Relation: $\text{Threshold Energy} = \text{Activation Energy} + \text{Average Kinetic Energy of Reactants}$. **Q20.** The hydration enthalpy of Cu^{2+} is much more negative than that of Cu^+, which more than compensates for the high second ionization enthalpy of copper. Thus, $2Cu^+(aq) \rightarrow Cu^{2+}(aq) + Cu(s)$. **Q21.** A is Acetone / Propanone ($CH_3 - CO - CH_3$) (formed by Markovnikov hydration of propyne). B is Sodium acetate / Sodium ethanoate (CH_3COONa) (formed via the Iodoform reaction).

Q22. (i) In chlorobenzene, the lone pair on chlorine resonates with the benzene ring, giving the $C - Cl$ bond a partial double bond character, shortening it. In CH_3Cl, it is a pure single bond. (ii) A racemic mixture contains equal proportions of both dextro (+) and laevo (-) enantiomers; the optical rotation of one is perfectly cancelled by the equal and opposite rotation of the other. (iii) Grignard reagents are extremely strong bases and will instantly react with water (moisture) to form alkanes ($RMgX + H_2O \rightarrow RH + Mg(OH)X$). **Q23.** (i) Propan-1-ol ($CH_3CH_2CH_2OH$) (Anti-Markovnikov addition of water). (ii) 2,4,6-Trinitrophenol (Picric acid). (iii) 2-Methoxytoluene (minor) and 4-Methoxytoluene (major). **Q24.** Moles of $MgCl_2 = \frac{1.9}{95} = 0.02\ mol$. Molality $m = \frac{0.02}{50/1000} = 0.4\ m$. Observed $\Delta T_f = 1.95\ K$. Calculated $\Delta T_f = K_f \times m = 1.86 \times 0.4 = 0.744\ K$. Van't Hoff factor $i = \frac{1.95}{0.744} \approx 2.62$. For $MgCl_2 \rightarrow Mg^{2+} + 2Cl^-, n = 3$. Degree of dissociation $\alpha = \frac{i-1}{n-1} = \frac{2.62-1}{3-1} = \frac{1.62}{2} = 0.81$ (or 81%). **Q25.** $\Lambda_m^\circ = \lambda^\circ(H^+) + \lambda^\circ(HCOO^-) = 349.6 + 54.6 = 404.2\ S\ cm^2\ mol^{-1}$. Degree of dissociation $\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{46.1}{404.2} = 0.114$. Dissociation constant $K_a = \frac{c\alpha^2}{1-\alpha} = \frac{0.025 \times (0.114)^2}{1-0.114} \approx 3.67 \times 10^{-4}\ mol\ L^{-1}$. **Q26.** (i) Draw an octahedral shape with Co in the center. *cis*-isomer: two Cl^- ligands are adjacent (90°). *trans*-isomer: two Cl^- ligands are opposite (180°). (ii) The *cis*-isomer is optically active because it lacks any plane of symmetry and its mirror image is non-superimposable. The *trans*-isomer has a plane of symmetry and is optically inactive. **Q27.** (i) Tollen's test: Benzaldehyde gives a silver mirror, Propanal also gives it. *Correction*: Use Fehling's test. Propanal gives a red precipitate, Benzaldehyde (aromatic) does not. (ii) Neutral $FeCl_3$ test: Phenol gives a violet colouration; cyclohexanol does not. (iii) Iodoform test: Ethanol gives a yellow precipitate of CHI_3 with $I_2/NaOH$; Diethyl ether does not. **Q28.** (i) $Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$. (ii) $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$. (iii) $2MnO_4^- + H_2O + I^- \rightarrow 2MnO_2 + IO_3^- + 2OH^-$.

Q29. (a) s^{-1} (or $min^{-1}, time^{-1}$). (b) After n half-lives, fraction left $= (1/2)^n$. For $n = 3$, fraction left $= (1/2)^3 = 1/8$ (or 12.5%). (c) 20% complete means 80% is left unreacted. $k = \frac{2.303}{t} \log\left(\frac{100}{80}\right) = \frac{2.303}{10} \log\left(\frac{10}{8}\right) = 0.2303 \times (\log 10 - \log 8)$. $k = 0.2303 \times (1.000 - 0.903) = 0.2303 \times 0.097 = 0.0223\ min^{-1}$. **Q30.** (a) Methanol (CH_3OH) and tert-Butyl iodide ($(CH_3)_3C - I$). (The reaction proceeds via S_N1 mechanism forming the highly stable $3°$ carbocation which is attacked by I^-). (b) Bromobenzene involves a very strong $C - Br$ bond with partial double bond character due to resonance. Hence, the phenyl halide strongly resists S_N2 nucleophilic substitution by an ethoxide ion. (c) Reactants: Bromomethane (methyl bromide, CH_3Br) and Sodium tert-butoxide ($(CH_3)_3C - O^-Na^+$). If roles are reversed (using tert-butyl bromide and sodium methoxide), the bulky strong base methoxide will cause elimination (E2) rather than substitution, yielding isobutylene (alkene) as the major product instead of the ether.

Q31. (a) Kohlrausch's law states that the limiting molar conductivity of an electrolyte is equal to the sum of the individual contributions of the limiting molar conductivities of its constituent cations and anions. $\Lambda_m^\circ(Al_2(SO_4)_3) = 2\lambda^\circ(Al^{3+}) + 3\lambda^\circ(SO_4^{2-})$. (b) Anode: $Fe \rightarrow Fe^{2+} + 2e^-$. Cathode: $2H^+ + 2e^- \rightarrow H_2$. Overall: $Fe + 2H^+ \rightarrow Fe^{2+} + H_2$. $n = 2$. $E_{cell}^\circ = E_{cathode}^\circ - E_{anode}^\circ = 0.00 - (-0.44) = +0.44\ V$. $E_{cell} = E_{cell}^\circ - \frac{0.0591}{2} \log \frac{[Fe^{2+}]}{[H^+]^2} = 0.44 - 0.0295 \log \frac{0.001}{1^2}$. $E_{cell} = 0.44 - 0.0295 \times \log(10^{-3}) = 0.44 - 0.0295(-3) = 0.44 + 0.0885 = 0.5285\ V$. **Q32.** (a) (1) They can exhibit multiple oxidation states to form unstable reaction intermediates. (2) They provide a large surface area with free valencies to adsorb reactants, lowering the activation energy. (b) Ni is $3d^8 4s^2$. Ni^{2+} is $3d^8$. Cl^- is a weak field ligand, so no electron pairing occurs in the 3d orbitals. The outer $4s$ and three $4p$ orbitals hybridize. Hybridization: sp^3. Geometry: Tetrahedral. Magnetic behaviour: Paramagnetic (due to the presence of 2 unpaired electrons in the 3d subshell). **Q33.** (a) Since oxidation of the alcohol **C** yields the exact acid **B**, they must contain the same carbon skeleton structure. The ester **A** ($C_8H_{16}O_2$) must be completely symmetrical, splitting into a 4-carbon acid and a 4-carbon alcohol. **B**: Butanoic acid ($CH_3CH_2CH_2COOH$). **C**: Butan-1-ol ($CH_3CH_2CH_2CH_2OH$). **A**: Butyl butanoate ($CH_3CH_2CH_2COOCH_2CH_2CH_2CH_3$).

Hydrolysis:

$CH_3CH_2CH_2COOCH_2CH_2CH_2CH_3 + H_2O \xrightarrow{H^+} CH_3CH_2CH_2COOH + CH_3CH_2CH_2CH_2OH$. Oxidation: $CH_3CH_2CH_2CH_2OH \xrightarrow{CrO_3,\, H^+} CH_3CH_2CH_2COOH$. (b) (i) HVZ Reaction: Carboxylic acids with an α-hydrogen react with Cl_2 or Br_2 in the presence of red phosphorus to form α-halo carboxylic acids. Example: $CH_3COOH + Cl_2 \xrightarrow{Red\ P} Cl - CH_2 - COOH + HCl$. (ii) Decarboxylation: Sodium salts of carboxylic acids when strongly heated with soda lime ($NaOH + CaO$) lose carbon dioxide to form an alkane with one carbon less. Example: $CH_3COONa + NaOH \xrightarrow{CaO,\, \Delta} CH_4 \uparrow + Na_2CO_3$.

CBSE Class 12 (2027) — Half-Yearly Predicted Paper (80% syllabus). AI-generated via PYQ/Blueprint analysis, not actual exam questions.